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Subject: PL Roll No.: 32 Exp. No.:
Name of the Experiment:

Performed on:
Submitted on:

Marks	Teacher's Signature with date
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Q1) List the key features and advantages of Python. Also mention any two Python IDEs or editors

Ans) • Key features :-

i) Simple and easy to learn - python uses English - like syntax, making it beginner friendly

ii) Interpreted Language - code is executed line by line which helps in easy debugging

iii) High Level Language - no need to manage memory manually

iv) Object Oriented - support classes, objects & inheritance

v) Portable - python programs can run on different platforms

• Advantages of Python :-

i) Faster program development

ii) Reduced Code Length

iii) Easy integration with other languages

iv) Widely used in web development

• Python Editors :-

i) IDLE ii) PyCharm

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Q2) Explain the history and evolution of python? Why is python popular?

Ans) i) Python was developed by Guido van Rossum in 199
ii) It was inspired by the ABC programming language.
iii) Python was designed with a focus on code readibility and simplicity
iv) Major versions include :-
→ Python 2.x – introduced many features but later discontinued
→ Python 3.x – improved performance, unicode support and security

- Reason for python popularity :-
i) Easy to learn for beginners
ii) Used in real world applications
iii) Large community support
iv) Rich libraries and frameworks
v) High demand in the industry

Q3) Python program for arithmetic operations

Ans)
```
a = float(input("Enter first no:"))
b = float(input("Enter sec no:"))
print("Addition", a+b)
print("Subtraction", a-b)
print("Multiplication", a*b)
print("Division", a/b)
```

- Explanation :- i) input() accepts value from user
ii) float() converts input into numeric type
iii) Arithmetic operators perform calculations

Q4) Analyze the code and identify data types and output
Example Code :-

```
x = 10
y = 3.5
z = "Python"
a = True
```

Ans) Data types :-

x → Integer (int)
y → Floating point (float)
z → String (str)
a → Boolean (bool)

- Output Explanation :- a) Each variable stores a different data type
b) Python automatically assigns data type based on value

Q5) Evaluate relational, logical and bitwise operators
Ans) Relational operators :- used for comparison
eg :-

```
a = 10
b = 5
print (a > b)
```

Logical operators :- used to combine conditions
eg :-

```
x = True
y = False
print( x and y)
```

Bitwise operators :- used for bit level operations

Eg :- a = 5
b = 3
print (a & b)

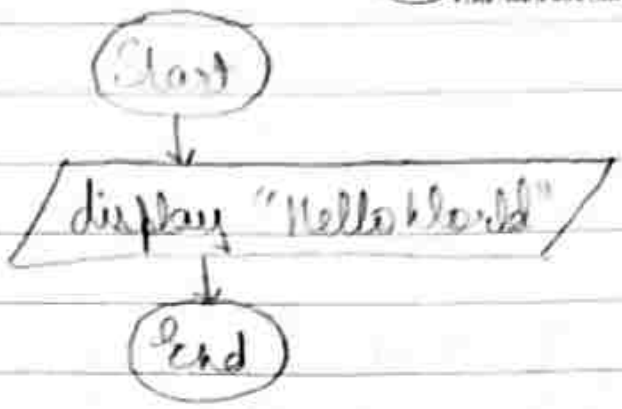
Q6) Menu Driven Python program.

Ans)
```
a = int (input ("Enter first no:"))
b = int (input ("Enter sec no:"))

print ("1. Arithmetic")
print ("2. Relational")
print ("3. Logical")
print ("4. Assignment")

choice = int(input("Enter choice:"))
if choice == 1:
    print ("Sum=", a+b)
elif choice == 2:
    print ("a >b =", a >b)
elif choice == 3:
    print ((a>0) and (b>0))
elif choice == 4:
    a += b
    print ("updated value of a =", a)
else:
    print ("Invalid choice")
```

Q7) Program to display "Hello World"

Ans) • Algorithm :-

i) Start
ii) Print Message
iii) Stop



• Program :-

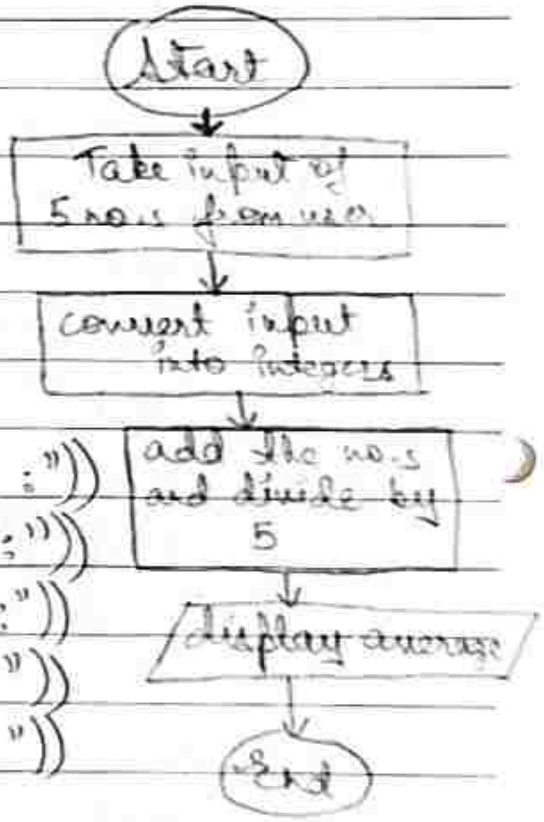
```
print ("HelloWorld")
```

• O/P :- HelloWorld

Q8) Program to find average of five numbers

Ans) • Algorithm :-

i) Accept five numbers
ii) Convert to integer
iii) Calculate average
iv) Display result



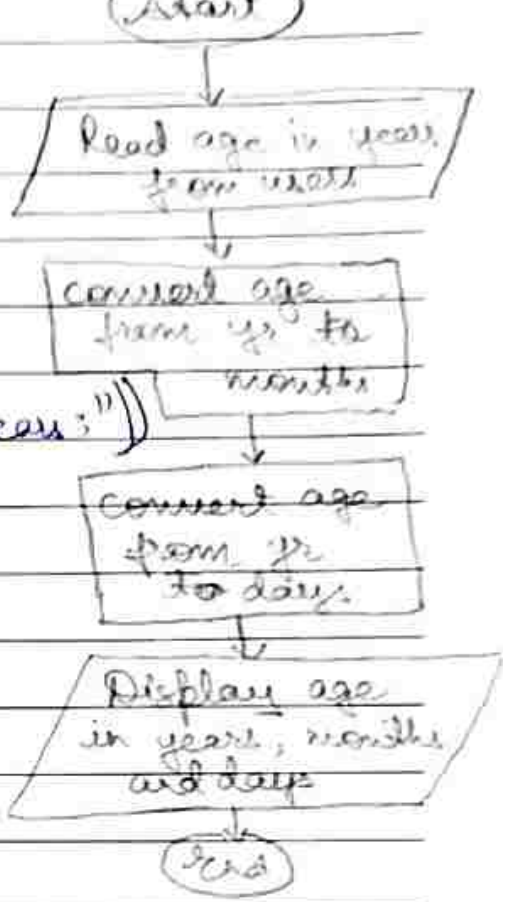
• Program :-

```
a = int (input ("Enter no. 1 :"))
b = int (input ("Enter no. 2 :"))
c = int (input ("Enter no. 3 :"))
d = int (input ("Enter no. 4 :"))
e = int (input ("Enter no. 5 :"))

avg = (a + b + c + d + e) / 5
print ("Average =", avg)
```

Q9) Program to convert age into years, months & days

Ans) • Algorithm :-

i) Input age in years
ii) Convert to months and days
iii) Display results



• Program :-

```
age = int (input ("Enter age in years:"))
months = age * 12
days = age * 365
print( "Age in years :", age)
print( "Age in months :", months)
print( "Age in days :", days)
```

Q10) Program to demonstrate data types and conversion

Ans) • Program :-

```
a = 10
b = 3.5
c = "25"
print (type(a))
print (type(b))
print (type(c))

c = int (c)
print ("After conversion :", type (c))
```